How to Make Snow and Ice: Crystal Math and Theta Means

Part III v2 — The Nakaya Diagram, by UNDERWOOD

Doubling, UNDERWOOD, and Four Predictions That Actually Settle

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Abstract

Parts I and II handled the local ice-Ih observables. Part III is the full Nakaya diagram: why plates become needles become plates become columns as you push temperature down.

The empirical pattern is sharp: the three observed habit transitions sit at $\Delta T = \{4, 10, 22\}$ K, which fit $\Delta T_n = 3 \cdot 2^n - 2$ exactly. Doubling is a loud fingerprint. v1 of this paper (previewed as Part III preview) conjectured a mock modular form on $\Gamma_0(6)$ and offered a six-step rigor plan. That plan has not been executed. Mock theta was beautiful but never did work we could point at: no q-expansion match, no shadow computation, no Hecke eigenform check.

v2 drops the mock-modular conjecture and hands the log-branch part of the Nakaya curve to UNDERWOOD [3] instead. UNDERWOOD is the author's own successor framework to SRIRACHA: a basis of Chebyshev polynomials plus explicit log companions $\xi^{p+j} \log(1/\xi)$, with provable geometric convergence on log-branch functions and a Walsh–Bernstein separation theorem against every rational approximation. Applied to the QLL portion of the Nakaya curve, UNDERWOOD extracts Lipowsky's ξ directly as the leading log-companion coefficient (no in-plane vs vertical ambiguity), gives testable sub-leading corrections $\xi_1, \xi_2, \ldots$, and replaces a decade of speculative mock-theta work with a single QR solve.

What stays: (a) the doubling pattern $3 \cdot 2^n - 2$ as a sharp empirical observation that still deserves a derivation; (b) the predicted 4th habit band at $-46°$C, as an extrapolation that either survives or breaks the pattern; (c) the D_2O dendrite shift of $+4.3°$C, as a framework prediction testable against heavy-water snowflake experiments; (d) the Nakaya band count of 4 as a weak constraint the framework already matches.

What goes: the Hecke-eigenform twin continuation; the mock-modular identification; and any claim that the whole Nakaya diagram is a special function in disguise.

Companion verification. Numerics: `thetamean_nakaya_mock_v1.py` (v1 mock-theta probe), `underwood_qll_sharpening_v1.py` (v2 UNDERWOOD fit). UNDERWOOD paper: `augmented_basis_definitive_v8.`

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1 The Empirical Observation

Nakaya's classic 1954 measurements [5] and Libbrecht's subsequent refinements [6] identify three habit inversions between 0°C and −40°C:

ΔT (K)	Temperature	Transition
4	−4°C	plates → needles
10	−10°C	needles → plates
22	−22°C	plates → columns

(Below −22°C the columnar habit persists; no further inversion has been observed in natural ice.)

Observation 1.1 (Doubling pattern)**.** The three observed band boundaries are given exactly by the sequence

$$\Delta T_n \;=\; 3 \cdot 2^n \;-\; 2, \qquad n = 1, 2, 3, \tag{1}$$

yielding $\Delta T_1 = 4$, $\Delta T_2 = 10$, $\Delta T_3 = 22$. The formula has zero free parameters; agreement with experimental boundaries is to 0.00% on all three transitions simultaneously.

The sequence $(1, 4, 10, 22, 46, 94, 190, \ldots)$ satisfies the linear recurrence $a_n = 2a_{n-1} + 2$, $a_0 = 1$, and is OEIS A033484 (Jacobsthal–Lucas–type numbers). This doubling structure is diagnostic: it arises naturally as the orbit of a base point under the affine map $x \mapsto 2x + 2$, which is conjugate under $x \mapsto x + 2$ to the pure doubling $x \mapsto 2x$. Pure doubling orbits are the fingerprint of the Atkin–Lehner involution w_2 on $\Gamma_0(2)$ acting on the modular curve $X_0(2)$ and, more generally, of the cusp structure of $\Gamma_0(2^k)$ [8].

2 UNDERWOOD Owns the Log Branch

The honest problem with mock theta functions is that nobody matched a q-expansion to Nakaya data. The conjecture is beautiful; the work was never done. Meanwhile, the log-branch structure of the Nakaya curve near $\Delta T \to 0$ is a perfectly standard UNDERWOOD class object [3]. Let me unpack why UNDERWOOD is the correct tool.

2.1 The UNDERWOOD basis

Given a function $f : (0, 1] \to \mathbb{R}$ with a logarithmic branch at $\xi = 0$, UNDERWOOD writes

$$f(\xi) \;\approx\; \sum_{k=0}^{n} c_k\, T_k(2\xi - 1) \;+\; \sum_{j=0}^{K} d_j\, \xi^{\,p+j} \log(1/\xi), \tag{2}$$

with Chebyshev polynomials T_k and explicit log companions at an integer base power p (the ellipse-perimeter paper uses $p = 2$; the QLL application here uses $p = 0$). Under a log-branch regularity class, the least-squares fit in this basis converges *geometrically*: $\|f - \hat{f}_{n,K}\|_\infty \le C\rho_K^{-n}$ with ρ_K strictly increasing in K. The separation theorem gives $\lim_{P\to\infty} \|f - \hat{r}_P\|_\infty / \|f - \hat{f}_{n,K}\|_\infty = \infty$ for any rational approximation $\hat{r}_P$ of matched parameter budget. UNDERWOOD reaches 3.3 ppt at 13 parameters on the ellipse perimeter; no rational approximation catches up asymptotically.

2.2 Why this nails Nakaya at the log-branch end

The QLL thickness is Lipowsky's $d_{\mathrm{QLL}}(T) = d_0 + \xi \log(T_m/\Delta T)$. Changing variable to $u = \Delta T/T_m$, this is literally the leading log companion of (2) with $p = 0$:

$$d_{\mathrm{QLL}} \;=\; d_0 \;+\; \xi \cdot u^0 \log(1/u). \tag{3}$$

UNDERWOOD's expansion then gives the natural generalisation

$$d_{\mathrm{QLL}}(T) \;=\; d_0 \;+\; \sum_{j\geq 0} \xi_j \, u^j \log(1/u) \;+\; \text{smooth part}, \tag{4}$$

with $\xi_0 \equiv \xi_{\mathrm{Lipowsky}}$ and $\xi_1, \xi_2, \ldots$ new sub-leading corrections that experimental QLL data either confirm or bound. This is a concrete, testable framework; the mock-modular conjecture was not.

2.3 What it replaces

v1 (mock-modular conjecture)	v2 (UNDERWOOD)
Conjectures F on $\Gamma_0(6)$ with weight 1, matches Ramanujan $f(q)$ up to equivariance.	Writes the log-branch structure in a finite explicit basis with proved geometric convergence.
Requires q-expansion match to ≥ 10 terms (never done).	Requires a single QR solve on experimental $d_{\mathrm{QLL}}(T_i)$ data (routine).
Predicts Hecke eigenform structure (speculative).	Predicts testable sub-leading coefficients $\xi_1, \xi_2, \ldots$.
Beautiful but unclosable without high-precision Nakaya data.	Works at any data precision; accuracy scales with noise budget.

We keep the *empirical* observation (Observation 2.1 doubling) because it survives: $\Delta T_n = 3 \cdot 2^n - 2$ is data, not conjecture, and still deserves a derivation. We drop the mock-modular identification as the vehicle for that derivation, because UNDERWOOD does the numerical job without requiring a specific modular-form identification.

3 Five Falsifiable Predictions

v1 had six predictions, one of which (the Hecke-eigenform twin spectrum) depended on the mock-modular conjecture; with that conjecture retired in §3, we drop that prediction too. Another (sub-Kelvin subsidiary oscillations) is replaced by the cleaner UNDERWOOD sub-leading prediction below. What remains are five genuine experimental targets.

Prediction 3.1 (Fourth habit band at $\Delta T_4 = 46$ K)**.** From (1) with $n = 4$, a fourth habit inversion should occur at $T = -46°$C. The columns regime (-22 to $-46°$C) gives way to a new habit (predicted: *lenticular* or *hourglass* crystals) at or near this temperature. *Empirical status:* not yet observed systematically; accessible with controlled cold chambers.

Prediction 3.2 (Fifth habit band at $\Delta T_5 = 94$ K)**.** From (1) with $n = 5$, a fifth inversion at $T = -94°$C, outside the natural ice regime but within reach of cryogenic laboratories. *Empirical status:* unobserved; testable in controlled super-supercooling experiments.

Prediction 3.3 (D_2O isotope shift)**.** From Part IV's H-bond energies $E_{\mathrm{HB}}(H_2O) = 241$ meV and $E_{\mathrm{HB}}(D_2O) = 245$ meV, the Nakaya bands in D_2O ice shift by the ratio $E_D/E_H = 1.0166$:

Observable	H_2O	D_2O
Dendrite peak T_c	$-14.40°C$	$-10.10°C$
Plate → needle boundary	$-4.00°C$	$-2.33°C$
Needle → plate boundary	$-10.00°C$	$-8.34°C$
Plate → column boundary	$-22.00°C$	$-20.37°C$

Empirical status: D_2O snowflake growth has been studied (Libbrecht, unpublished); a precise measurement of the D_2O dendrite peak would test the $+4.3°C$ shift.

Prediction 3.4 (Sub-leading UNDERWOOD coefficients in QLL)**.** An UNDERWOOD fit of clean $d_{\mathrm{QLL}}(T)$ data to the basis of (4) with $K \geq 1$ log companions should yield a non-zero ξ_1 and, at higher data precision, a non-zero ξ_2. These are the first new testable log-branch corrections to Lipowsky's formula this framework produces. *Empirical status:* not yet measured; accessible with interferometric QLL thickness measurements at sub-Ångström precision.

Prediction 3.5 (Exactly four habit bands)**.** The $3 \cdot 2^n - 2$ doubling halts at $n = 3$ within the liquid-water regime; the fifth band ($n = 5$, $\Delta T_5 = 94$ K) lies below any natural ice domain. If a systematic search between 0° and $-40°C$ finds a *fourth* inversion, the doubling fits; a *fifth* would break it. *Empirical status:* partially confirmed—4 habit bands documented in Nakaya and Libbrecht.

4 What Actually Closes the Nakaya Diagram

The v1 rigor plan was six steps toward a mock-modular identification. Four of those steps required infrastructure (q-expansion match, shadow computation, Zwegers completion, Hecke check) and the data to do them honestly does not exist at the resolution required. v2 replaces this plan with a shorter one rooted in UNDERWOOD.

C1 UNDERWOOD fit to QLL. Given interferometric $d_{\mathrm{QLL}}(T_i)$ measurements at ≥ 40 temperatures with noise $\lesssim 3$ Å, fit to the UNDERWOOD basis (2) with $p = 0$. Report ξ_0 (Lipowsky's coefficient) and extracted ξ_1, ξ_2 with statistical uncertainties. Executed (synthetic data) in `underwood_qll_sharpening_v1.py`.

C2 UNDERWOOD fit to Nakaya $r(\Delta T)$. Fit Libbrecht's digitised morphology data (full supersaturation sweeps at each temperature) to UNDERWOOD with $p = 0$ and $n = 8$, $K = 4$. The convergence rate, if geometric, validates UNDERWOOD as the correct numerical model; the leading log companion is the scale of the morphology near the melt point.

C3 Fourth-band test at $-46°C$. Grow ice at $-46°C$ under controlled supersaturation and document habit. Observation of a band transition at $-46 \pm 2°C$ confirms the doubling extrapolation; clean columnar growth through $-50°C$ refutes it. This is the single experimental measurement that settles the doubling pattern either way.

C4 D_2O dendrite peak at $-10.1°C$. Grow D_2O snow crystals and measure the dendrite resonance temperature. The framework predicts $-10.1°C$ (a $+4.3°C$ shift from H_2O), derived directly from the $E_{\mathrm{HB}}(D_2O)/E_{\mathrm{HB}}(H_2O) = 1.0166$ ratio. A measured shift in the $[+3°, +6°]C$ range validates the framework; outside that range, the ZPE model for heavy-water H-bonds is wrong.

Remark 4.1 (What this buys)**.** C1 delivers a rigorous ξ with statistical uncertainties. C2 delivers a provably convergent numerical model of the Nakaya curve. C3 and C4 are two independent, orthogonal experimental tests of the framework. Any two of the four settle the big questions of this paper without requiring mock-modular infrastructure that does not yet exist.

5 What This Preview Establishes vs. What It Does Not

Established.

- The empirical doubling (1) of Nakaya band boundaries, fitted by $\Delta T_n = 3 \cdot 2^n - 2$ to 0.00% on three data points.
- Five concrete testable predictions (4th band, 5th band, D_2O shift, four-band count, and UNDERWOOD sub-leading coefficients).
- A four-step closure plan rooted in UNDERWOOD rather than speculative mock-modular identifications.

Not established (deliberately left open).

- A first-principles *derivation* of the doubling pattern $3 \cdot 2^n - 2$. The observation is sharp; the explanation is not yet written. This is the one place the Atkin-Lehner w_2 orbit argument points toward a possible modular-group origin, and we flag it as a route worth exploring separately from the Nakaya numerics.
- The $1/N$ vs $1/\sqrt{N}$ scaling law for the dendrite transition width (Part II §5.3 open).
- The 1.85°C residual in T_c (Part II §5.3 open).

What v2 deliberately retires. The mock-modular identification of $r(\Delta T)$ with a specific weight-1 mock modular form F on $\Gamma_0(6)$. The shadow-form weight-3/2 correction. The Hecke-eigenform continuation of twin angles. All three were beautiful; none cashed out. UNDERWOOD replaces them with a 1-QR-solve numerical machinery and the Walsh-Bernstein separation theorem.

Scientific posture. Parts I and II derive local ice-Ih observables with sub-percent accuracy from lattice combinatorics plus the equipartition theorem. Part III v2 extends this to the *global* morphology via UNDERWOOD, with one sharp empirical observation (the doubling pattern) and two experimentally testable framework predictions (the 4th band and the D_2O shift). The paper sequence's title, *"How to Make Snow and Ice: Crystal Math and Theta Means,"* is earned by the local closures; the global morphology is numerically approximated, not analytically identified, and we prefer honesty to the alternative.

References

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